ELEC 221 Lecture 12
The CT Fourier transform properties: convolution and multiplication

Tuesday 20 October 2026

Announcements

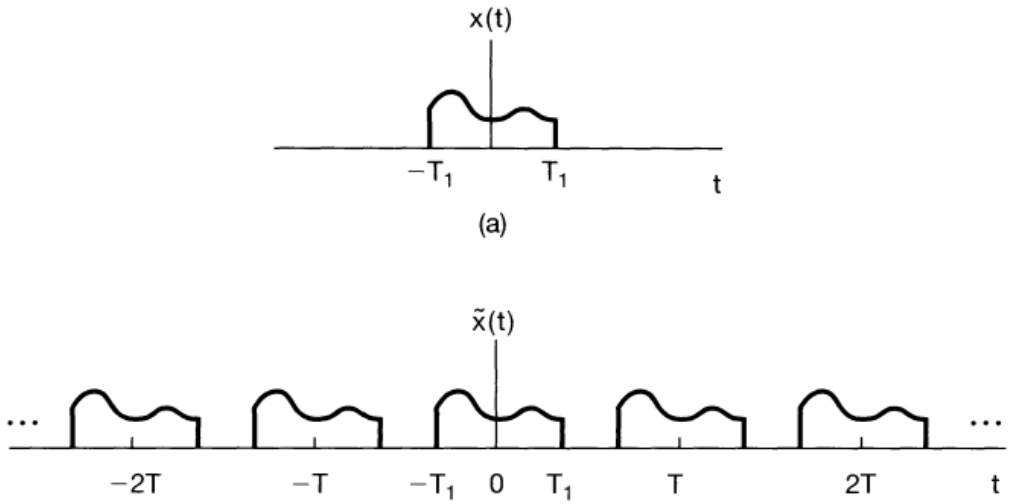
- Quiz 6 today
- Tutorial Assignment 3 due Friday 23:59

Exercise

If you finish quiz 6 early: compute the Fourier transform of $x(t) = e^{-at}u(t)$ $(a > 0)$.

Last time

We generalized the CT Fourier series (for periodic signals) to the Fourier transform (for aperiodic signals):



We expressed the periodic extension of an aperiodic function as

Image credit: Oppenheim chapter 4.1

Last time

We computed its Fourier coefficients:

We put this into the Fourier series and let $T \to \infty$ ($\omega \to 0$):

Last time

Inverse Fourier transform (synthesis equation):

Fourier transform (analysis equation):

Today

We will make a big step towards answering the question *"Why are we even doing this?"*

Learning outcomes:

- Compute the Fourier transform of periodic functions
- Describe the time/frequency duality of the Fourier transform
- Apply the *convolution property* of the Fourier transform to characterize LTI system behaviour
- Describe the *multiplication property* of the Fourier transform and provide applications of its use

Dirichlet conditions for Fourier transforms

If a signal

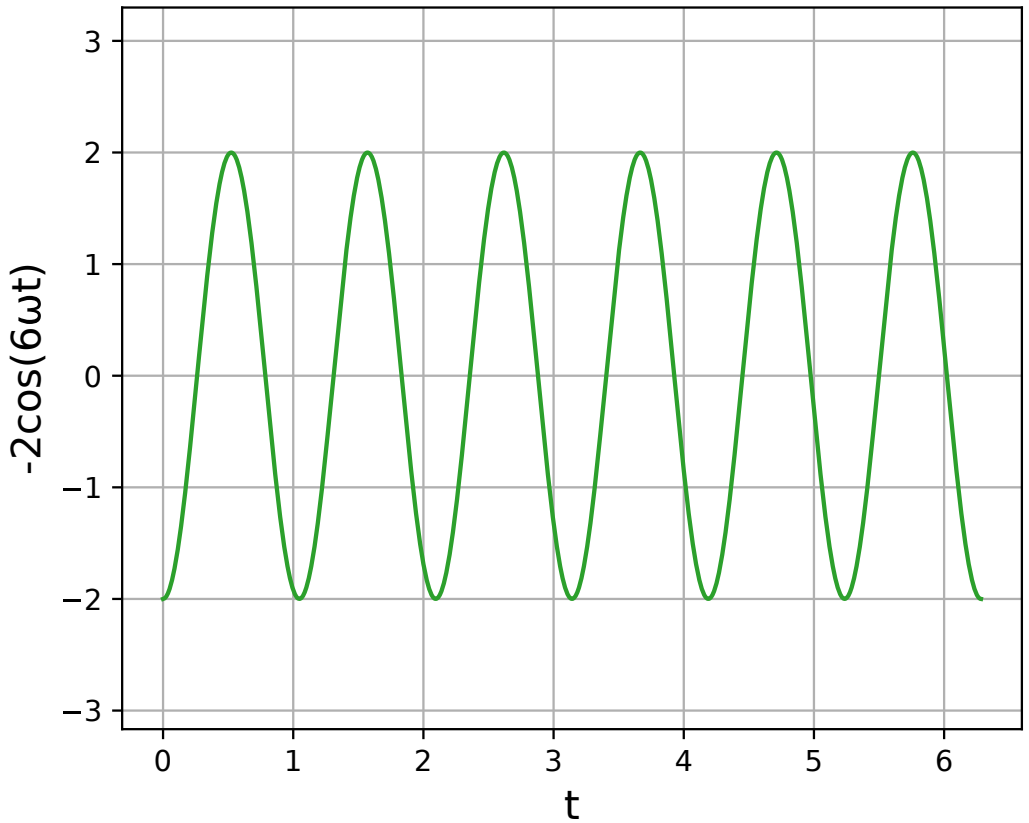
1. is single-valued
2. is absolutely integrable ($\int_{-\infty}^{\infty} |x(t)|dt < \infty$)
3. has a finite number of maxima and minima within any finite interval
4. has a finite number of finite discontinuities within any finite interval

then the Fourier transform converges to

- $x(t)$ where it is continuous
- the average of the values on either side at a discontinuity

Dirichlet conditions for Fourier transforms

The "**within any finite interval**" takes care of periodic signals:



Exercise 1

What signal does the following Fourier transform belong to?

$$X(j\omega) = 2\pi\delta(\omega - \omega_0)$$

Exercise 2

What signal does the following Fourier transform belong to?

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0)$$

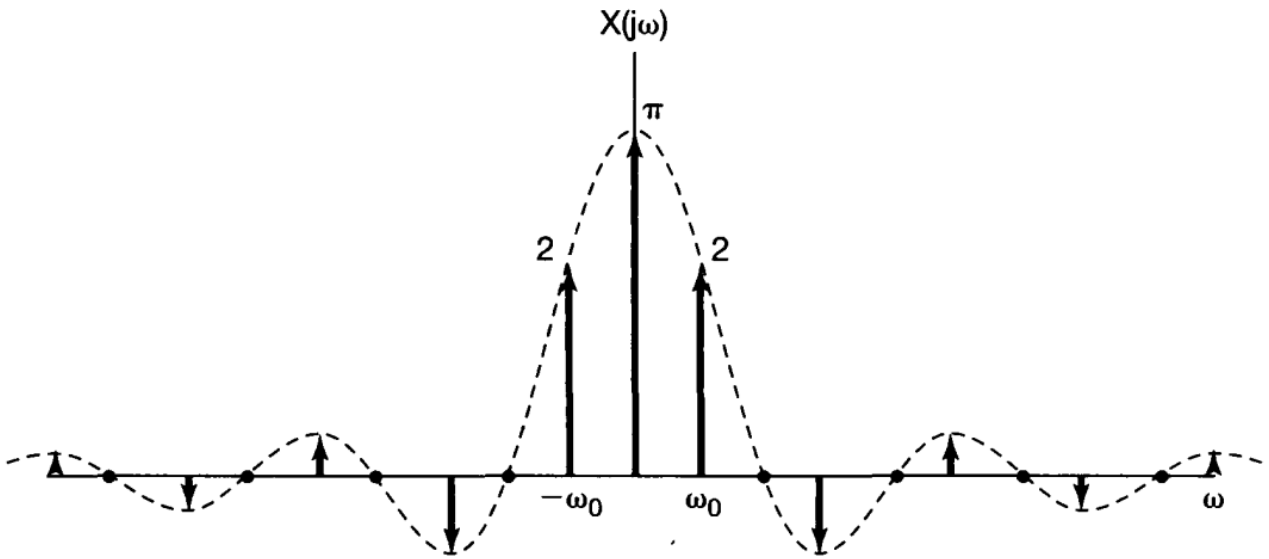
Fourier transforms for periodic signals: a unified representation

The Fourier transform of a periodic function is an **impulse train**.

The impulses have area $2\pi c_k$ and are positioned at the harmonically related frequencies.

Fourier transforms for periodic signals: a unified representation

Its Fourier *transform* will be



(Plotted for $T = 4T_1$)

Image credit: Oppenheim chapter 4.2

Fourier transform properties

I'll provide some notes / exercises to practice applying these properties.

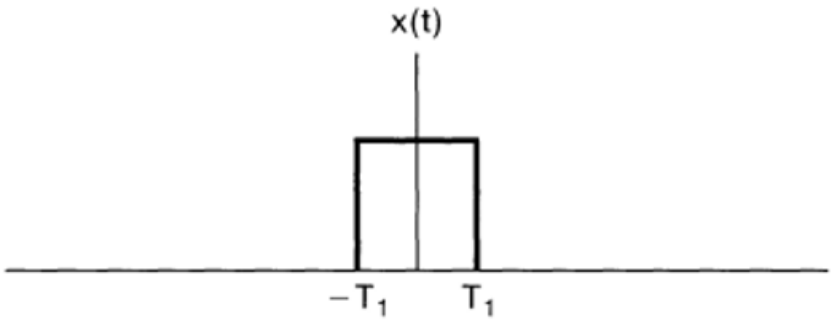
TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$ $y(t)$	$X(j\omega)$ $Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t}x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi}\int_{-\infty}^{+\infty}X(j\theta)Y(j(\omega-\theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^{t}x(t)dt$	$\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j\frac{d}{d\omega}X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re e\{X(j\omega)\} = \Re e\{X(-j\omega)\} \\ \Im m\{X(j\omega)\} = -\Im m\{X(-j\omega)\} \\ \lvert X(j\omega)\rvert = \lvert X(-j\omega)\rvert \\ \measuredangle X(j\omega) = -\measuredangle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}v\{x(t)\}$ [$x(t)$ real] $x_o(t) = \mathcal{O}d\{x(t)\}$ [$x(t)$ real]	$\Re e\{X(j\omega)\}$ $j\Im m\{X(j\omega)\}$

Image credit: Oppenheim chapter 4.6

Time/frequency duality of the FT

Let's consider a single square pulse:



Its Fourier spectrum is

Image credit: Oppenheim chapter 4.1

Time/frequency duality of the FT

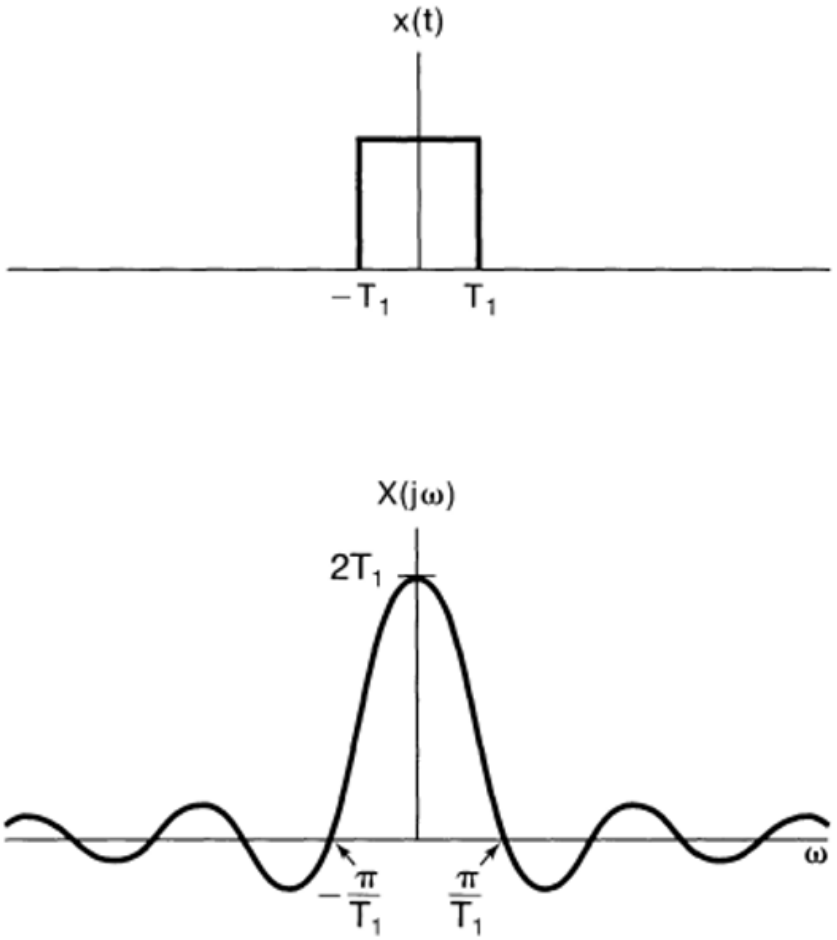


Image credit: Oppenheim chapter 4.1

Time/frequency duality of the FT

Now let's consider a signal whose Fourier transform is

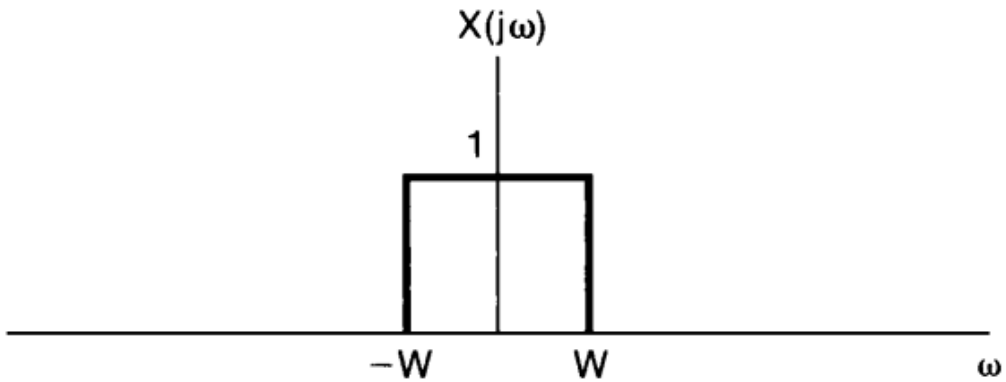


Image credit: Oppenheim chapter 4.1

Time/frequency duality of the FT

Compute the inverse Fourier transform:

Time/frequency duality of the FT

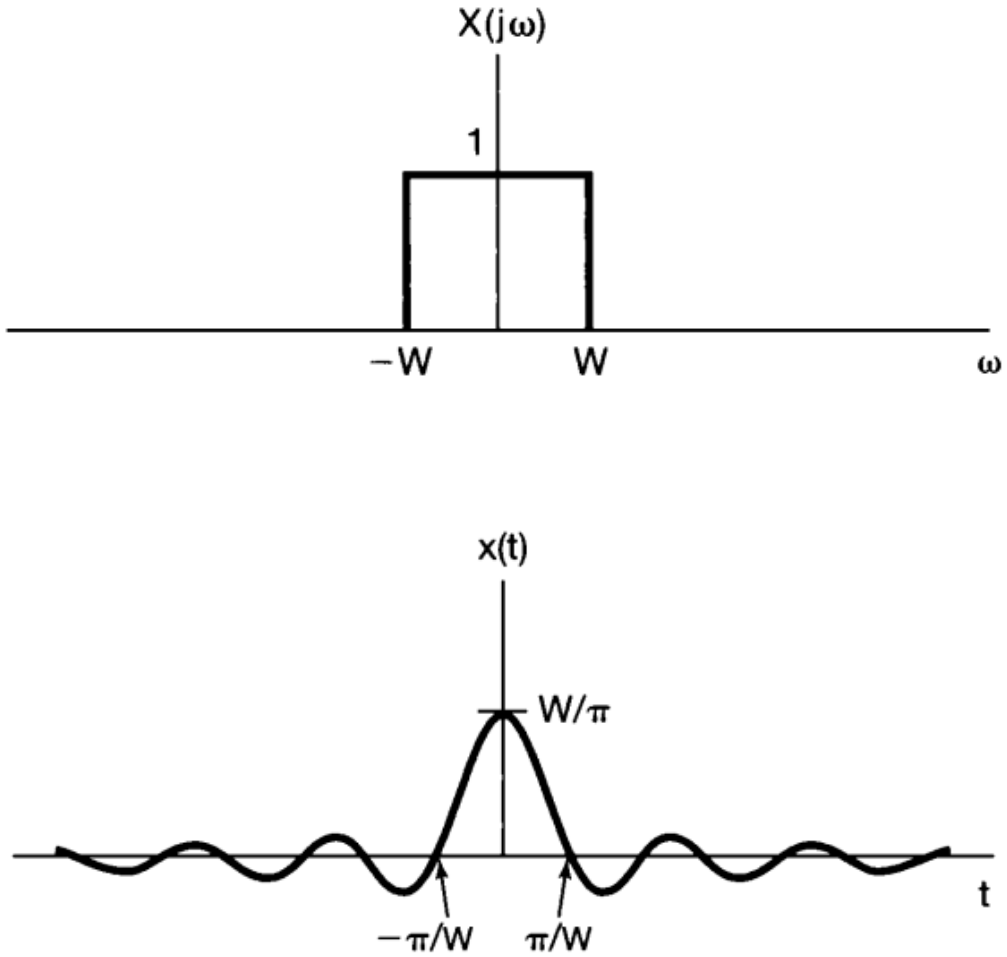


Image credit: Oppenheim chapter 4.1

Time/frequency duality of the FT

Duality: for any transform pair ($x(t) \leftrightarrow X(j\omega)$), there is a *dual pair* with the time and frequency variables interchanged.

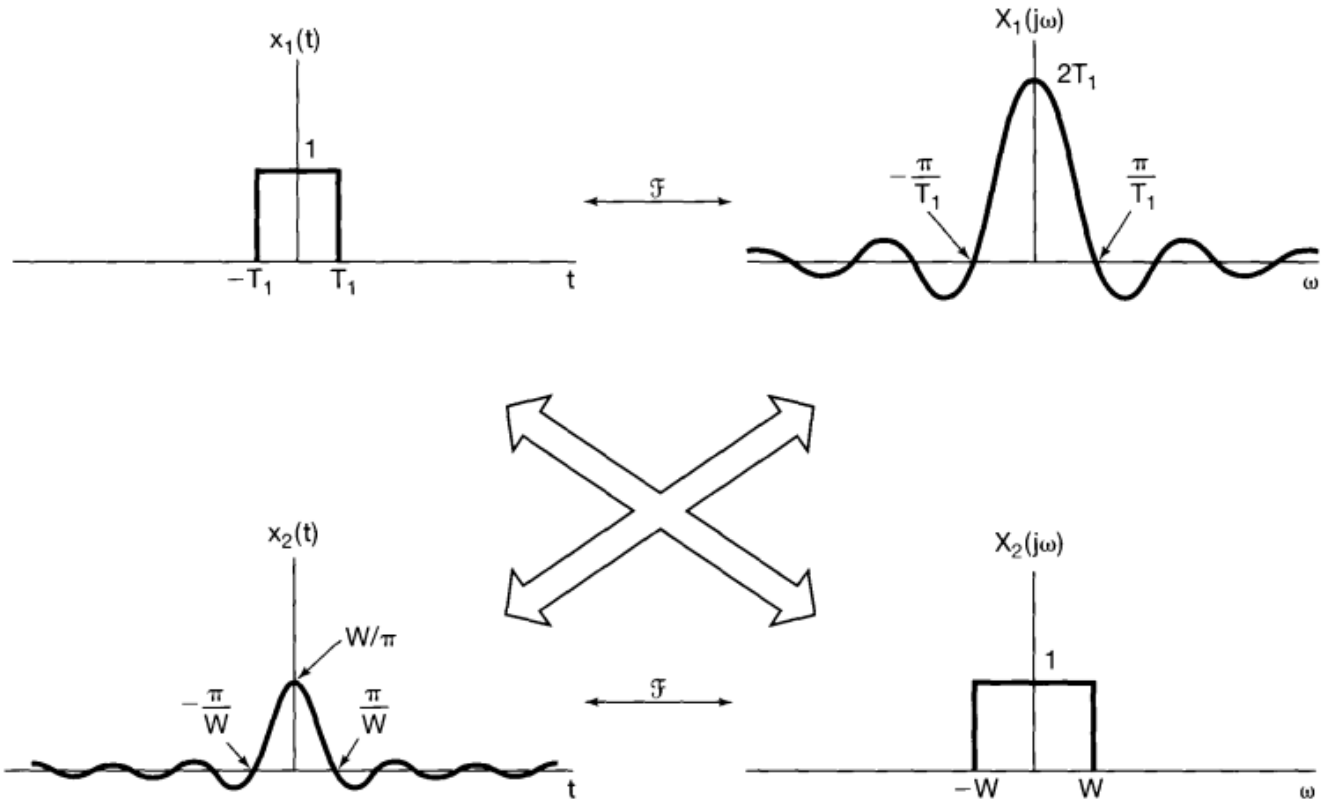


Image credit: Oppenheim chapter 4.3

Convolution and the Fourier transform

Recall complex exponentials are eigenfunctions of LTI systems. If we input signal $x(t)$ into LTI system with impulse response $h(t)$

where

This came from the convolution integral:

Convolution and the Fourier transform

Let's express $x(t)$ using the inverse Fourier transform:

and put this into the convolution integral...

Convolution and the Fourier transform

We have **two** ways to write $y(t)$:

This has an important implication:

Example: convolution

This can be helpful for evaluating the output of systems given $h(t)$ and $x(t)$ (or $h(t)$ given $y(t)$ and $x(t)$, etc.)

Example: suppose a signal $x(t) = \frac{\sin(\omega_0 t)}{\pi t}$ is input into a lowpass filter with frequency response

Method 1: inverse FT $H(j\omega)$ to get $h(t)$, then convolve.

Example: convolution

Method 2: compute $X(j\omega)$ then use convolution property.

We just computed $h(t)$ and found

Exercise: convolution

Consider an LTI system that sends

What is its impulse response?

Exercise: convolution

The multiplication property

We know that:

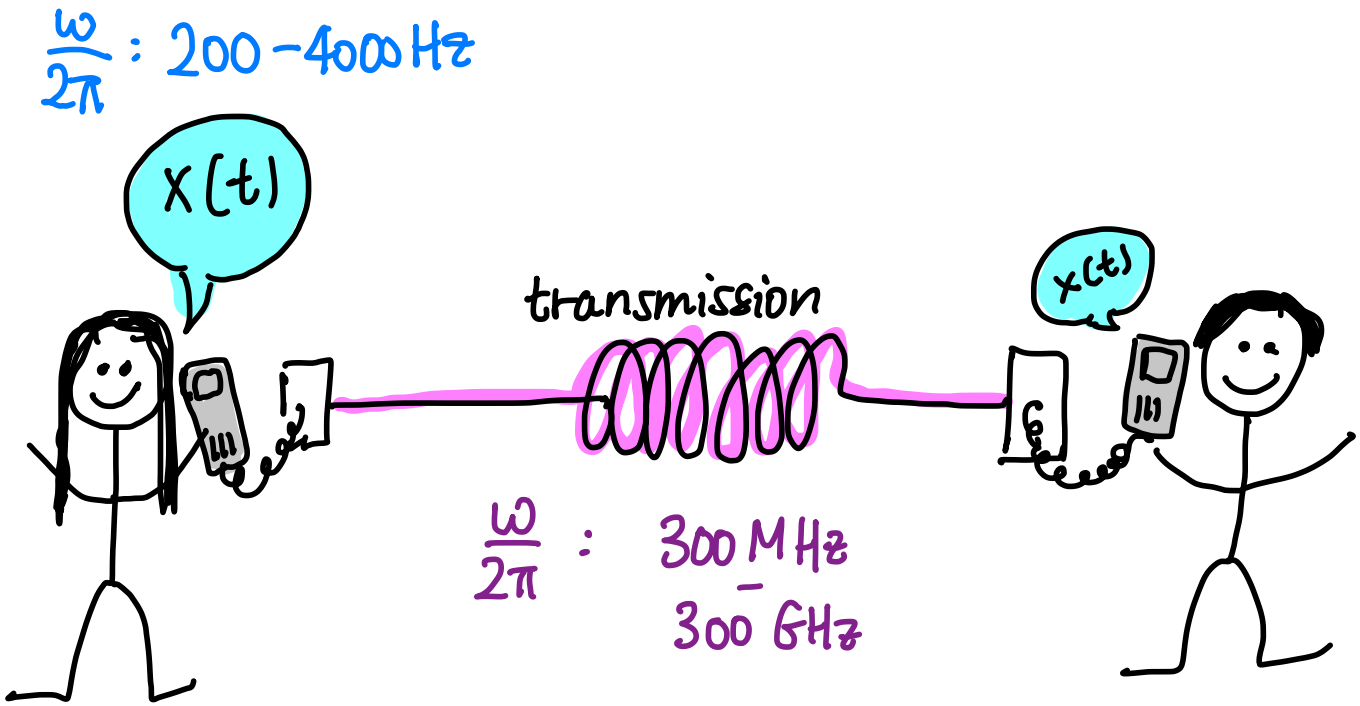
Something similar holds when we interchange time and frequency:

This is the **multiplication property**.

Example: the multiplication property

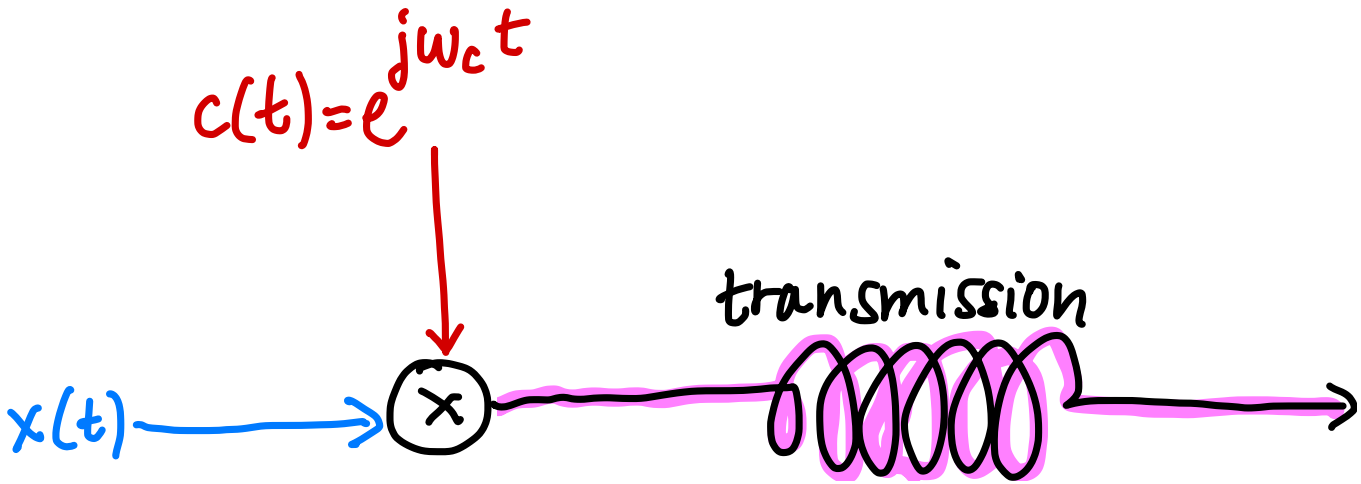
We will explore this more in depth when we discuss communication systems and signal **modulation**.

For now, here is a taste:



Example: the multiplication property

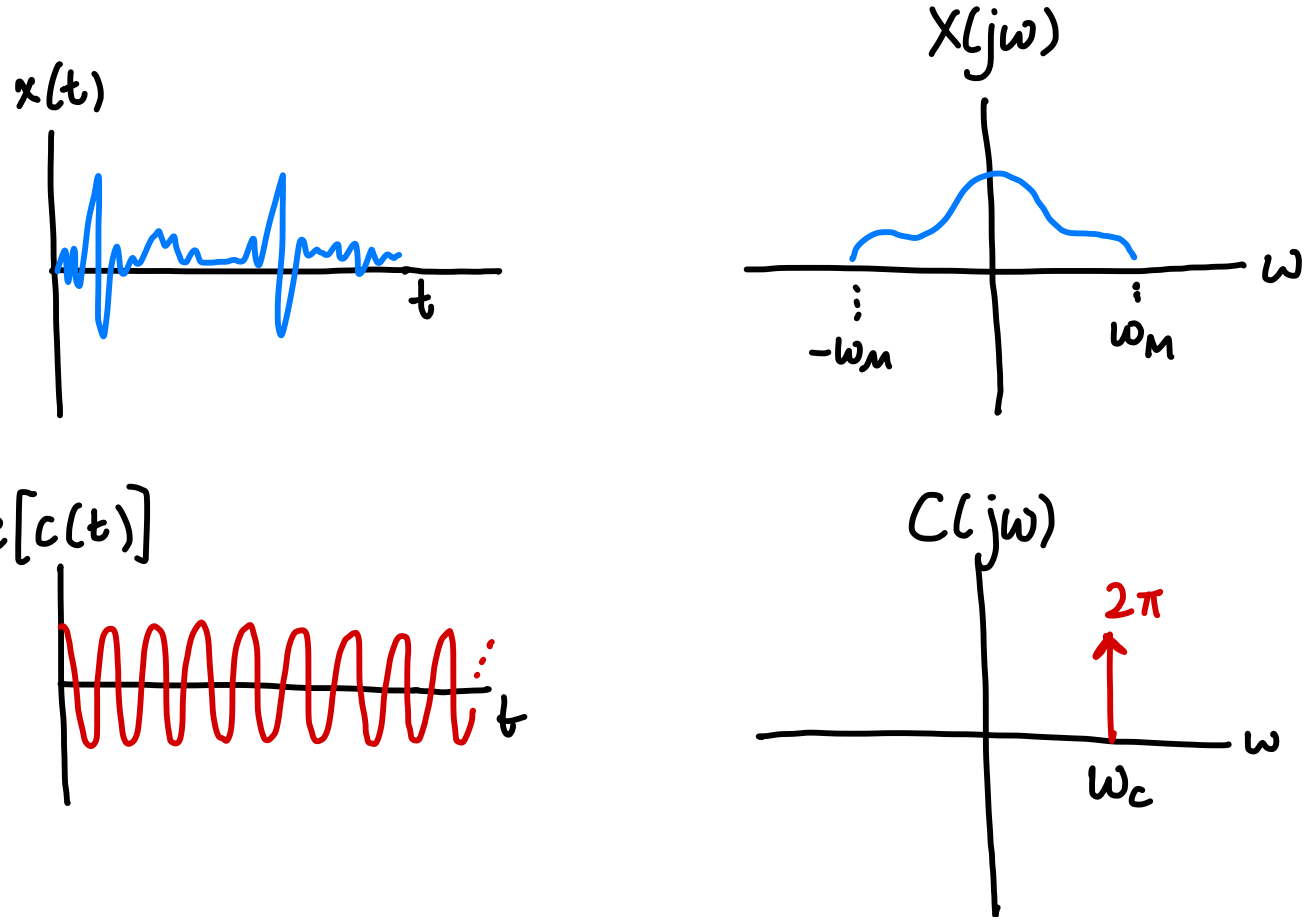
To shift our signal into the frequency range of transmission, we can multiply it by a **carrier signal** (amplitude modulation):



Is this doing what we think it is?

Example: the multiplication property

Consider the Fourier spectrum of both signals:



Example: the multiplication property

The multiplication property tells us

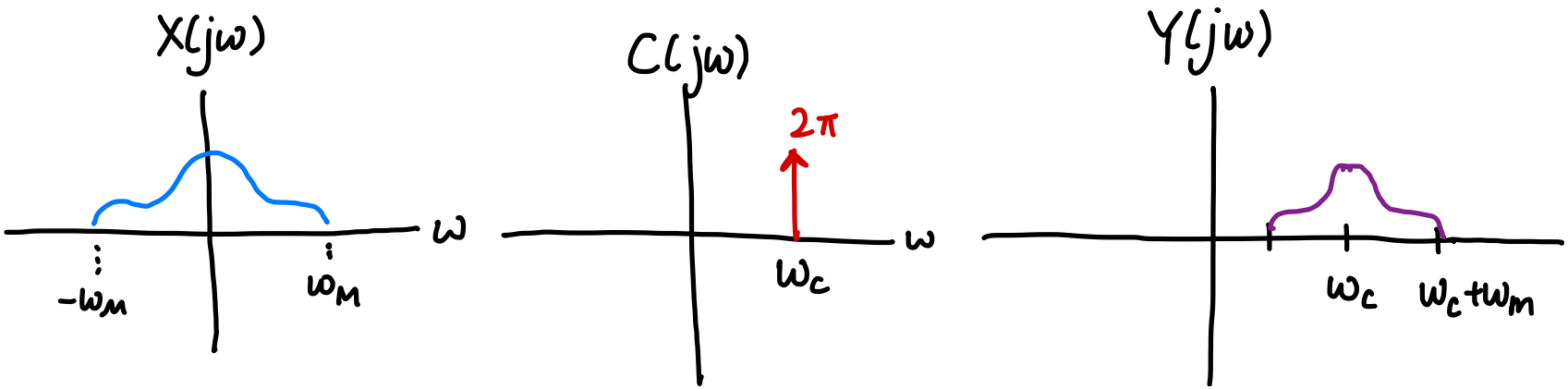
We have

Example: the multiplication property

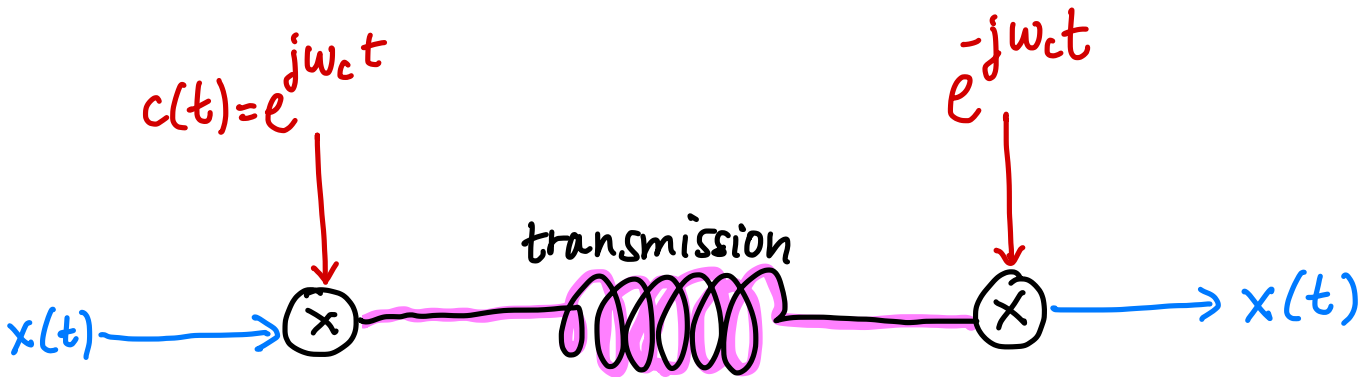
Let's convolve them:

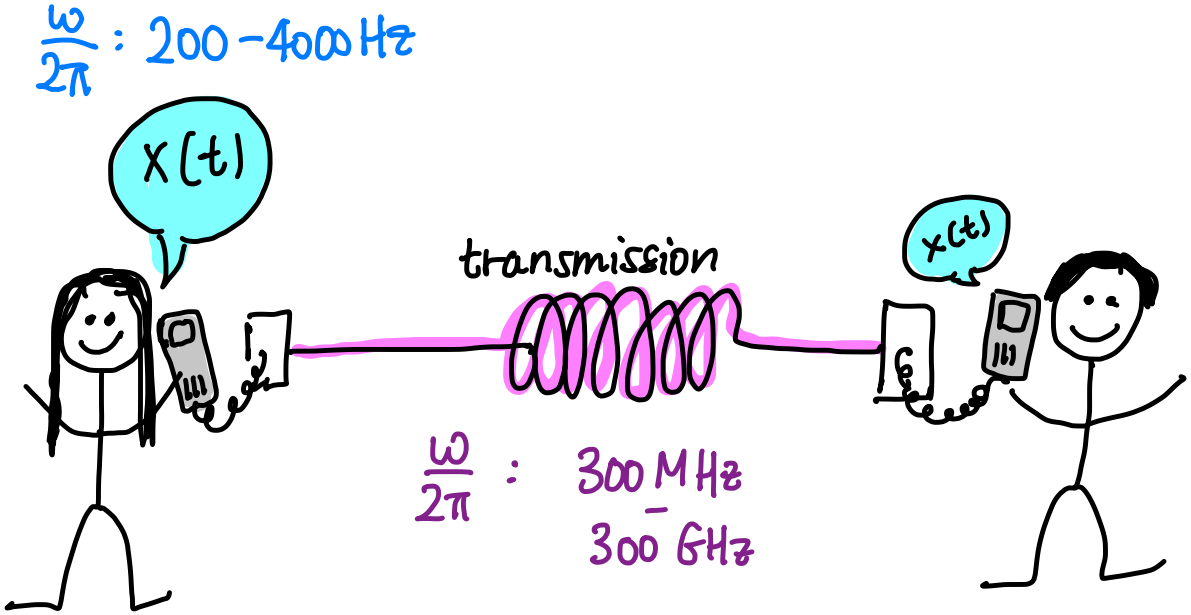
Example: the multiplication property

Multiplication with complex exponential carrier signal shifts the spectrum. We can move it into the desired frequency range.



Example: the multiplication property





Example: frequency-selective filtering with variable centre frequency

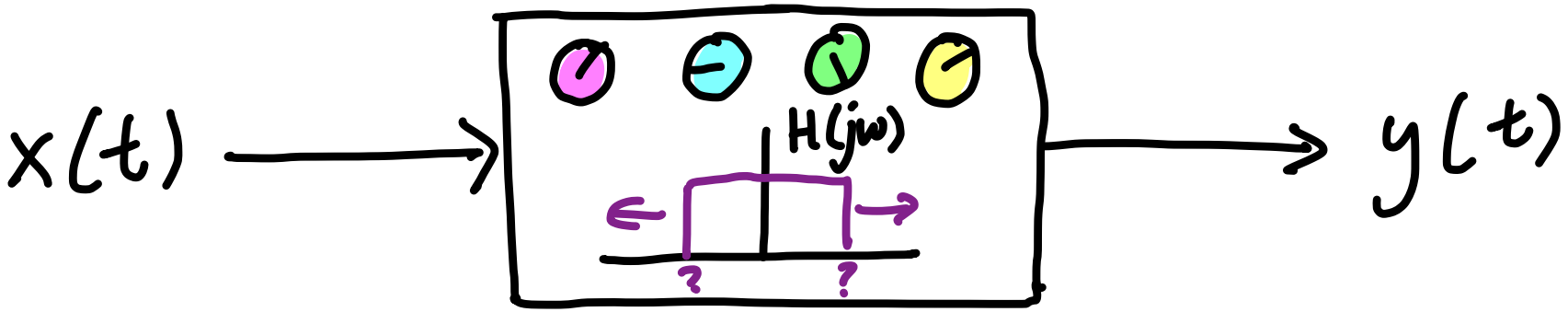
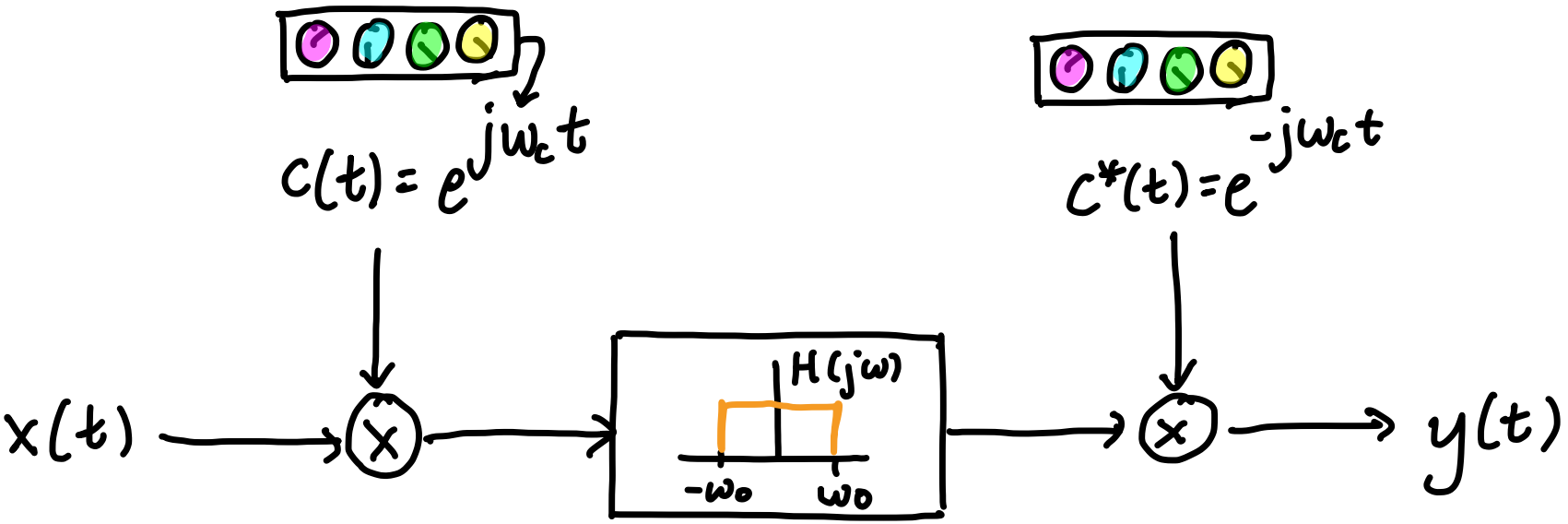


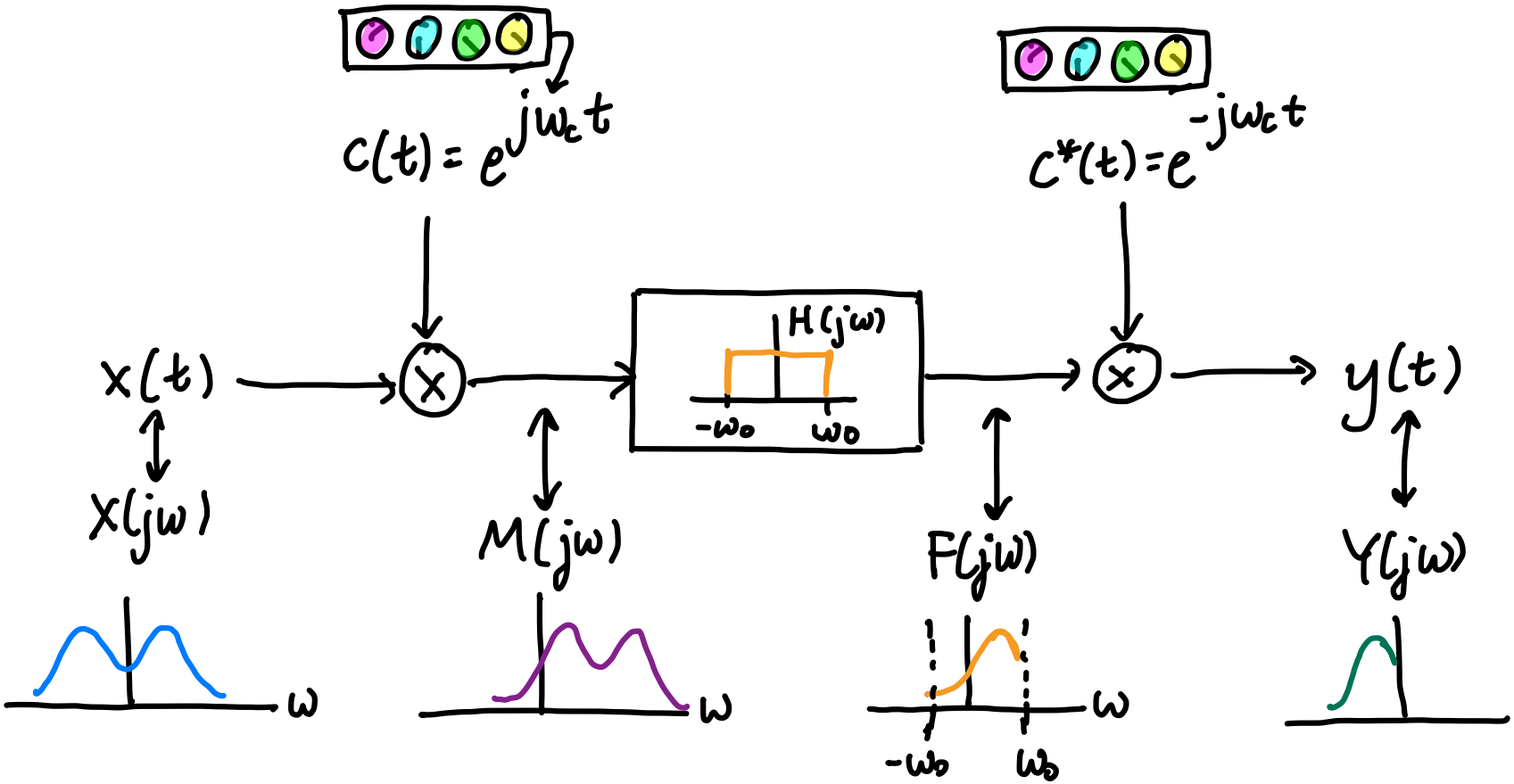


Image: https://www.euronics.ee/UserFiles/Products/Images/162391-panasonic-rf-2400d-radio.png

Example: frequency-selective filtering with variable centre frequency



Example: frequency-selective filtering with variable centre frequency



Real-world example: radio



"Superheterodyne receiver"

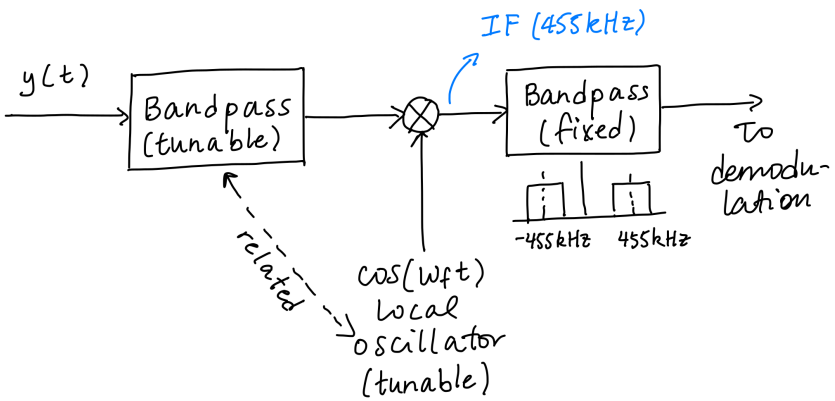


Image: https://www.euronics.ee/UserFiles/Products/Images/162391-panasonic-rf-2400d-radio.png

For next time

Content:

- Behaviour of the Fourier transform under differentiation and integration
- LTI systems based on differential equations

Action items:

1. Tutorial Assignment 3 due Friday 23:59

Recommended reading:

- From today's class: Oppenheim 4.4-4.6
- Suggested problems: 4.4, 4.6, 4.9, 4.12, 4.15, 4.17, 4.19, 4.26, 4.32
- For next class: Oppenheim chapter 4.7, 6.1-6.2